MCGILL-FIAM ASSET MANAGEMENT HACKATHON 2026 · FINALS

Mont Royal Carlo presents

Market-neutral · US equities · out of sample Jan 2021 – Aug 2026

Mont Royal Carlo · Alex Zhao · Faig Haji · Oliver Zhou · Bohan Zhang · Tony Tran

SECTION 1

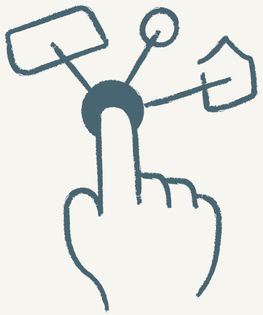
Executive summary

What we harvest. Stock-selection alpha: returns from picking the right stocks within their peers, not from size, market or sector bets. Minimizing exposure from size, market, and sector bet, while trying to generate alpha from stock selection,.

How we forecast. Gradient-boosted trees on 147 firm characteristics rank US stocks on next-month return, trained only on the past. Each month the forecast is stripped of size, market beta and sector.

How we build the book. An optimizer holds 100 longs and 100 shorts at 175% gross: dollar- and beta-neutral, every sector within ±2% net, size exposure within ±0.02.

What it returned. Over 68 out-of-sample months (Jan 2021 – Aug 2026), $1bn grew to $2.99bn after 20 bp trading costs, with no losing year.



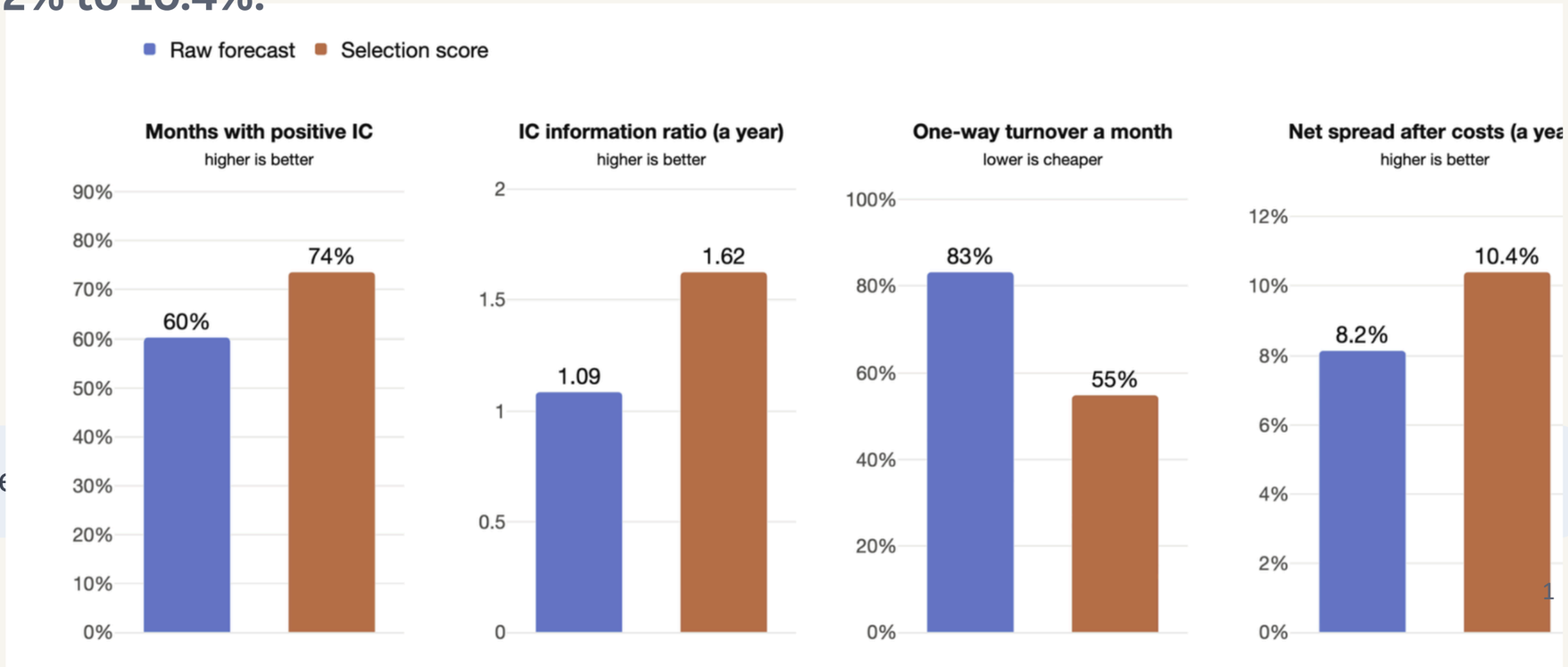
SECTION 2

From predictions to tradable signals

A non-linear engine (LightGBM) reads 147 firm characteristics. These three carry the most weight:

The result: Our selection score delivers more consistent predictions

- **Reduces monthly turnover by 34%,**
- **and increases annualized net long-short spreads from 8.2% to 10.4%.**



Before the optimizer: the score

SECTION 3

Data and methodology



Data

Used: 147 firm characteristics, monthly, 2015 – Aug 2026 (about 529,000 stock-months).

Excluded, 8-K filings: they cover 93.2% of firms still listed but 1.6% of firms that left, so a missing filing leaks the future.

Carbon Arc used for testing our models

Model and validation

Model: gradient-boosted trees (LightGBM), no 8-K inputs.

Target: next-month excess return, demeaned within each month, so it learns who beats whom.

Inputs: ranked to [–1, 1] each month; missing values kept as missing, since they carry information.

Splits: expanding training window, 2-year validation, refit yearly. Design 2019–20, test Jan 2021 – Aug 2026.



SECTION 4

Portfolio construction



1 **Screen.** Price ≥ $5, no nano or micro caps, shorts only above $500m market cap at formation.

2 **Build the pools.** Longs: the 100 best-ranked names. Shorts: a pool of the 1,000 worst-ranked names.

3 **Pick 100 shorts from the pool.** The optimizer chooses the shorts and all weights: maximise the forecast minus a trading-cost penalty and a risk penalty.

4 **Hard constraints.** Gross 175%, net 0, beta 0, each name 0.1–2%, each sector's net within ±2%, size exposure within ±0.02, predicted vol ≤ 12%.

Why a 1,000-name short pool: the 100 worst names cluster in a few sectors and in small caps. A wide pool lets the shorts match the longs on sector and size.

Rebalance monthly: the signal comes from slow-moving characteristics, so trading faster would mostly pay costs.

SECTION 5

Risk agent evidence



Position level

Every name between 0.1% and 2% of capital; the risk penalty keeps volatile names small.

Only stocks above $5; shorts only above $500m market cap at formation.

Book level

Dollar- and beta-neutral, sector net ±2%, size ±0.02, predicted vol ≤ 12%, gross 175%.

Sector agent vs fixed ±2% band

Each month the agent sets the sector band and a cap on each sector's share of gross, from masked inputs. Rule written first; 68 decisions logged.

Concentration: Health Care 25.4% → 21.1% of gross; largest sector 27.0% → 22.1%.

Net IR: 0.79 → 0.69 under Tony's constraints, just inside our 0.10 limit. Under our original constraints, 1.01 → 0.74: failed.

Example: April 2024, it cut Health Care from 32.5% to 20.2% and earned +1.58% vs +0.36%. Yet across 2024 it lagged 0.64% a month.

Verdict: less concentration costs return. Tested on our earlier optimizer; the final book keeps the fixed ±2% band.

SECTION 6 · EXHIBIT 1

Performance, Jan 2021 – Aug 2026

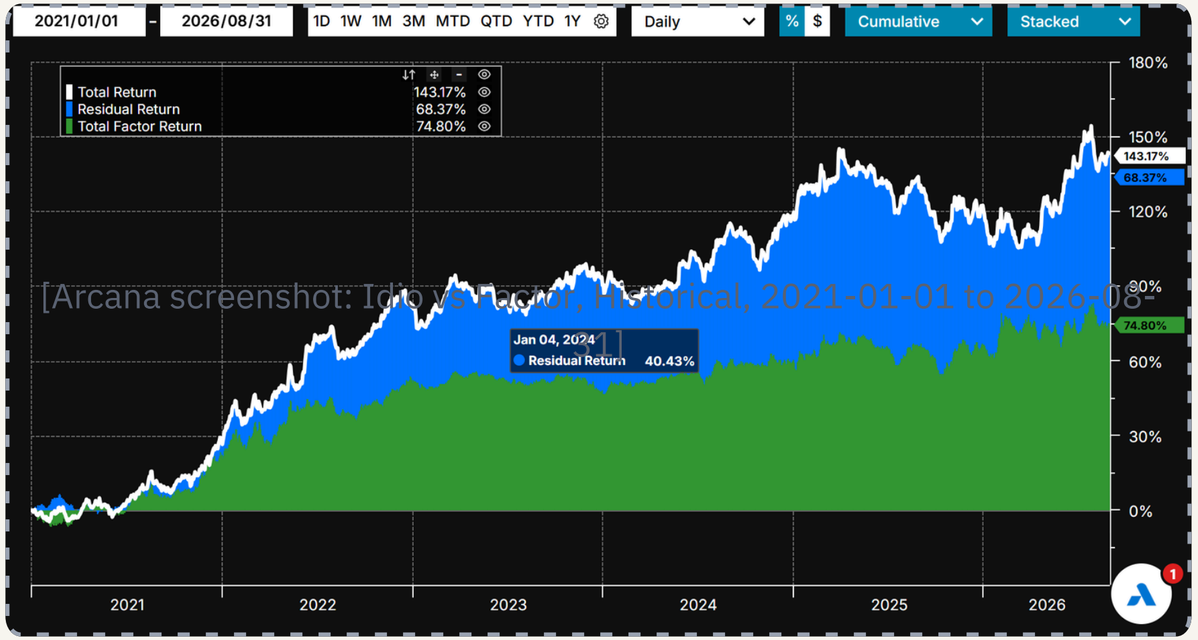
Metric	Strategy	Benchmark
Ending value ($bn)	2.99	1.51
Cumulative return	199%	50.9%
Ann. return, arith. / CAGR	20.3% / 21.3%	7.5%
Annualised volatility	13.1%	0.6%
Beta, monthly (s.e.)	−0.07 (0.11)	n/a
Sharpe ratio	1.29	n/a
IR vs T-bill + 4%	0.98	n/a
Tracking error	13.2%	n/a
Max drawdown, monthly / daily	−7.1% / −16.2%	0%
Max 1-month loss	−6.7%	none

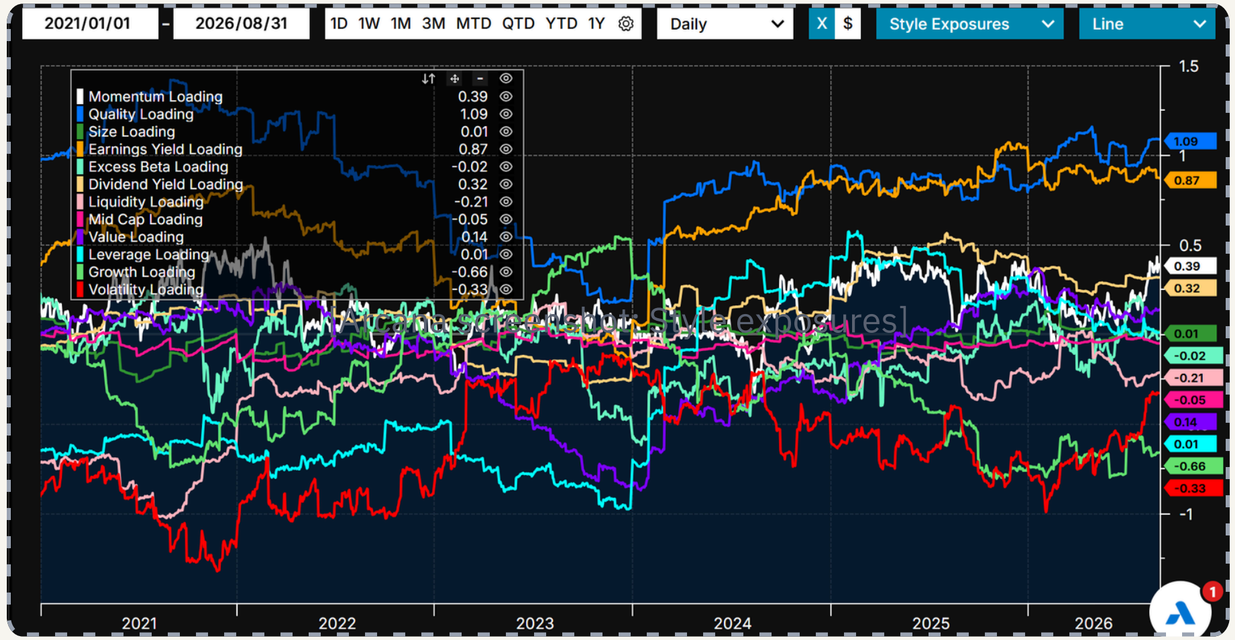
Metric	Strategy	Benchmark
Positions avg / min / max	200 / 200 / 200	n/a
Gross avg / max, % cap.	175 / 175	n/a
Net avg / min / max, % cap.	0 / 0 / 0	n/a
Turnover, per rebal. / ann.	25% / 302%	n/a
Avg position size, $m	16.8	n/a
Short book median cap, $m	2,554	n/a
Trading cost, % NAV/yr	1.25 at 20 bp	n/a
Borrow cost, % NAV/yr	0.9 at 1%	n/a
IR net of all costs	0.70	n/a

Monthly stats ×12, ×√12. Net of 20 bp unless noted; all costs = 65 bp + 1% borrow. Beta vs value-weighted US market; daily drawdown from Arcana.

SECTION 7

Independent verification: Arcana





Selection vs factors: Arcana credits +68% of AUM to stock-specific return and +70% to style, with industry +6% and beta −1%. Both legs add stock-specific return: long +22%, short +47%.

Where we disagree: our regression puts 98% of the average return in alpha; Arcana gives style about half. Arcana predicts beta −0.22 on average; realised is −0.07 (s.e. 0.11). We report both.

Arcana: Historical, before costs, % of AUM, cumulative. It shows 196 names on average: a few just-acquired stocks can't be priced.

SECTION 8

Costs and implementation

What we charge

Base case: 20 bp per dollar traded.

Stress case: 65 bp, our measured square-root market impact, plus 1% a year to borrow the shorts.

Turnover: 25% one-way a month, about 3× a year. Trading costs 1.25% of NAV a year at 20 bp.

Cost assumption	IR
Before costs	1.08
20 bp per dollar traded	0.98
65 bp + 1% borrow	0.70

The trading-cost penalty sits inside the optimizer, so the book only trades when the forecast change is worth more than its cost.



IR vs T-bill + 4%, Jan 2021 – Aug 2026, monthly rebalancing; turnover = ½Σ|Δw|.

SECTION 9

Discussion

Behaved as designed

Stock selection carried the return; sector P&L was only 0.4% a year.

No losing year; both legs added stock-specific return in Arcana.

Where it fell short

2023 and 2025 were weak: IR 0.06 and 0.14.

Realised vol 13.1% against at most 9.4% predicted.

Style factor exposure are not neutralized.

With more time

A risk model whose predicted vol matches realised.

8-K filings pulled from EDGAR by company ID.

An agent for sector caps.

What we learned: most of our forecast's skill sits in stocks nobody can trade. We built the book where it can be traded, and reported what that costs.

APPENDIX A1

What we tried and set aside

Our first registered book: net IR 0.06 after harsh costs, and Health Care net up to +37%, which breaks the 20% sector cap. Rebuilt.

Unscreened optimizer: IR 1.12, but 40% of gross sat in stocks under $5. Not tradeable at $1bn.

Pre-declared retrained model (A): net IR 0.74, but 27% vol and a −48% drawdown. Breaks the vol cap.

Constraint search: about 50 settings tested on tradeable stocks; all lost money in 2019–20. We stopped and fixed the setup on principle.

8-K filings: no signal in three tests, and the file leaks survival. Excluded.

Carbon Arc as signals: foot traffic IC t = 1.45, card spend 1.56 with a near-zero spread, web traffic 2.54 but descriptive only. Not added.



APPENDIX A2

Sector band comparison



Sector band	IR	Vol	Max drawdown	Beta	Turnover
±2% (chosen)	1.43	15.7%	−12.5%	−0.11	343%
±5%	1.28	17.5%	−13.6%	−0.17	345%
±10%	1.08	18.5%	−14.8%	−0.18	358%
No limit	1.29	19.1%	−20.1%	−0.31	367%
Exactly 0%	n/a	n/a	n/a	n/a	no solution

We chose ±2% because it is the tightest band the optimizer can solve, not because it scored best. All four runs are reported.

Monthly research engine, flat 20 bp costs, before the hedge. Relative comparison only.

Mont Royal Carlo

thank you